Predicting an unknown endpoint

Exercise sheet 1

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Part I. Theory

1. A fixed but unknown endpoint

Let $L > 0$ be fixed but unknown, and let

$$U_1, \dots, U_N \overset{\text{i.i.d.}}{\sim} \text{Uniform}[0, L].$$

The task is to predict L from the entire vector $(U_1, \dots, U_N)$. Consider the three hypotheses

$$h_1(u) = \frac{2}{N} \sum_{i=1}^{N} u_i, \qquad h_2(u) = \max_{1 \le i \le N} u_i, \qquad h_3(u) = \frac{N+1}{N} \max_{1 \le i \le N} u_i.$$

(a) Formulate this as a regression problem. Specify feature and label spaces, the joint distribution of features and labels, and the loss function corresponding to mean squared error. Choose spaces that do not depend on the unknown L.

(b) Calculate the population risks $R_L(h_1)$, $R_L(h_2)$ and $R_L(h_3)$. Give their dependence on N and L, and compare them for $N = 10$.

2. A two-stage experiment

First draw $L \sim \text{Uniform}[1, 5]$. Given L, draw ten independent observations $U_1, \dots, U_{10} \sim \text{Uniform}[0, L]$. The task is to predict L from these ten observations.

(a) Specify feature and label spaces, the joint distribution, and the loss function for this regression problem. What changes compared with Task 1?

(b) Calculate and compare the population risks $R(h_1)$, $R(h_2)$ and $R(h_3)$ for this experiment, using $N = 10$.

(c) Write the empirical risk for m independently generated labelled examples. Distinguish m from the number of observations in one feature vector.

Part II. Practical work

(a) Generate 10,000 labelled training examples from Task 2. Train and compare networks with architectures

$$10 \to 16 \to 1, \quad 10 \to 64 \to 1, \quad 10 \to 32 \to 32 \to 1, \quad 10 \to 64 \to 64 \to 32 \to 1.$$

Use the notebook's common training settings and sorted input vectors.

(b) Compare the training and test errors of the four fitted hypotheses and h_1, h_2, h_3, using 20,000 common, independent test examples. Present a table and a plot. Compare with your theoretical results.

(c) For each fixed $L \in \{0.5, 1, 2, 3, 4, 5, 6, 8\}$, evaluate all seven hypotheses on 5,000 fresh feature vectors. Plot the empirical errors against L. Keep the trained networks unchanged. Discuss the results inside and outside $[1, 5]$.

Submission: Your theoretical solutions and a short discussion as a PDF, together with an executable notebook containing the numerical results and plots.